Performance Analysis of Filtered OFDM for 5G

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***Abstract*—Orthogonal Frequency Division Multiplexing (OFDM) is a pertinent multi-carrier modulation approach which is more immune to frequency selective fading. In 5G waveform, in order to reduce the traffic in OFDM based technology, it is significant to re-size the bandwidth. Consequently, a spectrally localized waveform technology called Filtered Orthogonal Frequency Division Multiplexing (F-OFDM), which is primarily an approach to sub-band based filtering is introduced. Each of the different sub-bands can be handled according to the traffic scenario. This work focusses on analyzing the performance of Filtered OFDM and OFDM in terms of Power Spectral Density (PSD) and Bit error rate (BER). The spectral efficiency in F-OFDM is increased by the reduction of out-of-band (OOB) emission. Simulation for the performance analysis of OFDM and F-OFDM in terms of PSD and BER have done in MATLAB. The working of OFDM has carried out in Verilog and the performance of all components is verified by giving different inputs.**

***Index Terms*—Orthogonal Frequency Division Multiplexing (OFDM), Filtered Orthogonal Frequency Division Multiplexing (F-OFDM), Power Spectral Density (PSD), Bit Error Rate (BER), Out Of Band (OOB), Signal to Noise Ratio (SNR), Inter-Symbol Interference (ISI), Inter-Carrier Interference (ICI), Cyclic Prefix (CP).**

I. Introduction

In recent wireless communication scheme, a multi-carrier approach is a promising waveform in 4G and Long Term Evolution (LTE). Orthogonal Frequency Division Multiplexing (OFDM) is a form of multi-carrier modulation [4] scheme which dissociate the data streams to transmit into many parallel data streams. OFDM adopts many sub-carriers that are closely spaced and are orthogonal to each other so that mutual interference can be eliminated. To maintain the total data rate which is similar to the typical single carrier modulation schemes in the same bandwidth, suitable modulation schemes have to be adopted. In order to superimpose these modulated data on the orthogonal sub-carriers, a series of orthogonal oscillators are used. Then, the output from the parallel oscillators is added up by a summer to produce an OFDM signal. These oscillators and the summer are replaced with an IFFT block which was recommended by Weinstein and Ebert [8,10], to scale down the complexity of OFDM [6]. Each OFDM symbol has pilot sub-carriers for the channel estimation [9]. OFDM has numerous disadvantages including sensitivity to drift and carrier frequency offset, high Peak Power to Average Ratio (PAPR), high OOB (Out of band) emission and so on. The PAPR can be reduced by filtering without reducing BER [7]. In order to meet the expectation of the Fifth generation (5G) [2,3] and overcome some of the problems encountered in OFDM, a new technology called Filtered OFDM (F-OFDM) is introduced.

The filtered OFDM divides a band into many sub-bands and for each sub-band, a multiple number of sub-carriers with different sub-carrier spacing is included. Since Filtered OFDM has sub-band and sub-carrier flexibility, different services can be accommodated for each sub-band. A well-designed filter with proper design has to be applied for the Filtered OFDM. Filtered OFDM is an approach of sub-band based filtering [1] and for each sub-bands, a filter with some specific properties have to be preferred. Since the bandwidth of the filter is larger than the sub-carrier filtering, the filter's impulse response length can be condensed. In F-OFDM, a well-designed filter is incorporated after adding the cyclic prefix in OFDM symbol. The length of the filter can go beyond the length of cyclic prefix [5]. In this work, we present three parts of simulation. Initially OFDM and functionalities is verified using Verilog. Second part gives the simulation study of spectral efficiency and OOB emission in F-OFDM. Last part of simulation includes BER performance of F-OFDM and OFDM. Simulation results illustrates that the performance of Filtered OFDM is preferable than that of Orthogonal Frequency Division Multiplexing (OFDM).

The paper is organized as given: Section II describes the architecture of both OFDM and F-OFDM. Section III describes the design of the filter in F-OFDM. Section IV shows the simulation results of OFDM and F-OFDM in terms of PSD and BER in MATLAB. Section IV also includes the simulation result of OFDM in verilog. Finally Section V concludes the paper.

II. Architecture of OFDM and F-OFDM

A. OFDM Architecture

OFDM scheme requires N number of sub-carriers to transmit the data streams. Each of these carriers is orthogonal to each other and centered at multiples of frequencies. These serial data streams are converted to N parallel data streams and then they are digitally modulated using appropriate modulation techniques like BPSK, QPSK, QAM and so on. The constellation mapper or Lookup Table is used for the purpose of modulation. For the superimposition of the modulated data on the orthogonal sub-carriers, it demands N sinusoidal oscillators tuned with N orthogonal frequencies that are parallel to each

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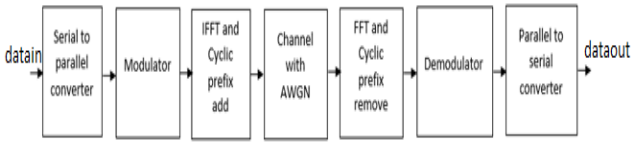


Fig. 1. OFDM Architecture.

other. The output of the sinusoidal oscillators is added up together that results to produce an OFDM signal. These oscillators and the summer are replaced with an IFFT block which was recommended by Weinstein and Ebert to scale down the complexity of OFDM. From the IFFT output, the OFDM symbol samples can be attained. The IFFT block switches the signal from frequency domain to time domain. The Fig. 1 shows the OFDM Architecture.

The Inter-symbol-Interference (ISI) imposes a negative impact on the OFDM which is induced by the delay spread. Delay spread occurs since multiple copies of the transmitted signals are received at different intervals of time than a single time. But ISI results when the delay spread goes beyond the symbol time duration. ISI can be eliminated by the use of the cyclic prefix. The cyclic prefix is a manner of adjoining some last portion of the OFDM symbol at the beginning of the OFDM symbol. The Inter-carrier-interference (ICI) can also be eliminated by the use of the cyclic prefix.

The channel adds AWGN (Additive White Gaussian noise) to the received signal. The reverse operation of transmitter side appears at the receiver side. At the receiver section, the transmitted signal is converted from analog to digital and then removes the cyclic prefix. The receiver has to perform synchronization (both channel timing and frequency), channel estimation, demodulation, and decoding. The output from FFT and the input of the IFFT are same. Finally, the original signal can be recovered by reassembling all the data streams from the individual carrier.

B. F-OFDM architecture

In F-OFDM, a band dissociates into a different number of sub-bands and sub-band filtering is executed for each sub-band. For, each sub-band, number of sub carriers, sub-carrier spacing and bandwidth are diverse. Also, each sub-band has the diverse number of the cyclic prefix, transmission time interval, IFFT size, and FFT size. Because of this sub-band and sub-carrier flexibility, different services can be accommodated for each sub-band. The Fig. 2 shows the architecture of F-OFDM.

Since sub-band based filtering is executed for independent OFDM systems, it offers inter-sub-band asynchronous transmission. In the transmission part, a well-designed filter is incorporated after adding the cyclic prefix in the time domain. Only sub-carriers close to the edges gets affected when the pass-band of the filter passes through the bandwidth of the

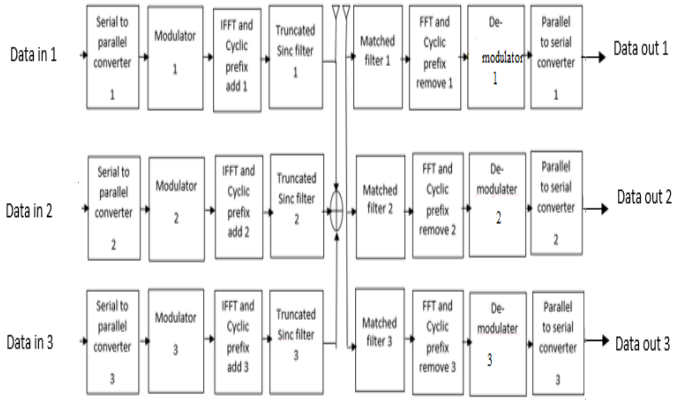


Fig. 2. F-OFDM architecture.

signal. The length of the filter is granted to exceed the length of the cyclic prefix. In filtered OFDM, the length of the filter is fixed at half the symbol duration. So the orthogonality of the sub-carriers gets vanished due to the design of the filter. This leads to the cutback of the out of band emission. Therefore, only few guard bands are admitted between the sub-bands which leads to asynchronous transmission. Since guard-bands or guard tones are included between the sub-bands, there is no interference of sub-bands. The guard band has spaced the multiplicity of the typical OFDM space. In order to enhance the utilization of spectrum, only few guard bands are included between the sub-bands.

Filtered OFDM lessens utilization of guard band which leads to the utilization of spectrum. For the vehicle transmission, the duration of transmission time interval lessens and enhances the spacing between sub-carriers for lessening latency and enhancing the reliability. After passing the signal through a noisy channel, it passes through a matched filter. A Matched filter is used in order to get the desired signal to noise ratio (SNR).

III. Design of the Filter in F-OFDM

The filter used for the filtered OFDM should follow some of the properties below for the proper filtering:

i) Flat passband is a prerequisite for the sub-carriers.
ii) In order to lessen the consumption of guard band, sharp transition is a prerequisite for the filter.
iii) Requires sufficient stop band attenuation.

The sinc filter or an ideal low pass filter shows all the above properties. This filter eliminates all the components of the frequency above a cut off frequency. The frequency response of the sinc filter is a rectangular function. The sinc filter impulse response is calculated by taking the inverse Fourier transform (IFT) of the filter's frequency response. The impulse

response of sinc filter is given by,

$$h(t) = \frac{2B \sin \pi(2Bt)}{\pi(2Bt)} = 2B\text{sinc}(2Bt) = \frac{1}{T_0}\text{sinc}(t/T_0).$$

The impulse response of the sinc filter is sinc function whose amplitude decreases as time increases. Then sample this function at the rate of $T_0 = (1/2B)$.

The sampling output is obtained by the multiplication of the signal with the periodic train of impulses which are separated by the spacing T_0.

The sinc filter output is the sum of the shifted impulse response and the transmitted signal at an interval of an integral multiple of T_0.

The impulse response has an infinite response in the positive and negative directions. For the real world applications, this impulse response has to be truncated with a specific window like Hanning window. The objective of the window is to convert the impulse response from infinite to finite response. There is a smooth transition of zero at both ends by window method. There exist some ripples in the output of the truncated sinc function. So in order to avoid these ripples, extend both ends of the low pass filter with tone offset. Tone offset is the additional number of sub-carriers added with the desired output.

At the receiver section, a matched filter is used which correlates a known signal with the unknown signal in order to recover the desired signal on the receiver side. The received signal is the desired signal plus AWGN. The objective of the matched filter is to detect the desired signal from the noisy received signal. In this case, there is a high noise power than the power of the signal and therefore the SNR value gets enhanced. In order to increase the SNR value, a matched filter is used in the Filtered OFDM at the receiver section. In a digital communication system, consider that a binary sequence which is coded in non-polar return to zero passes over a noisy channel. These sequences are described as unit pulse sequence and bit 1 can be represented by $+1$ and 0 can be represented by -1. Then the message signal received is the sum of shifted unit pulses.

IV. Simulation Results

In this section, the following blocks of OFDM are implemented and the final output is verified in Verilog. Also, it involves MATLAB simulation result of OFDM and Filtered OFDM in terms of the Power spectral density (PSD) and Bit error rate (BER). Then compare the simulation results of Filtered OFDM and OFDM which results that Filtered OFDM has better performance than OFDM.

A. Verilog Implementation of OFDM

1) Scrambler: It is necessary for the data to randomize before transmission. The scrambler eliminates the transmission of long sequences of zeros and ones and also eliminates unwanted regularity for the signal transmitted. This can be implemented with 15 width LFSR and two XOR gates. The simulation result of the scrambler is shown in Fig. 3.

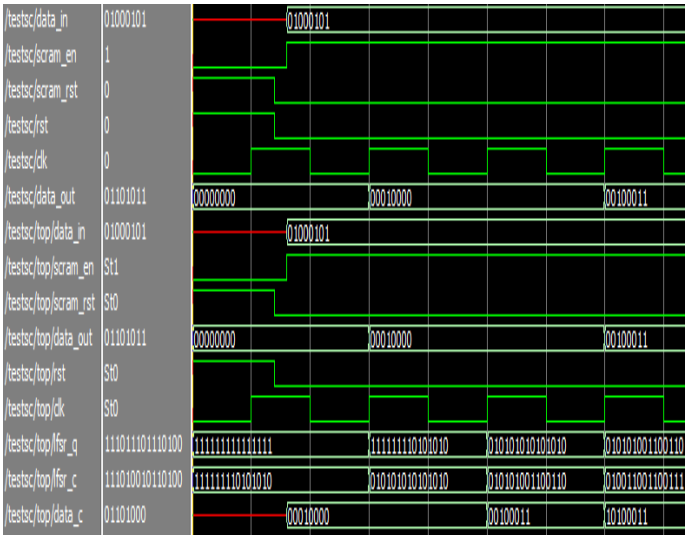


Fig. 3. Simulation result of scrambler.

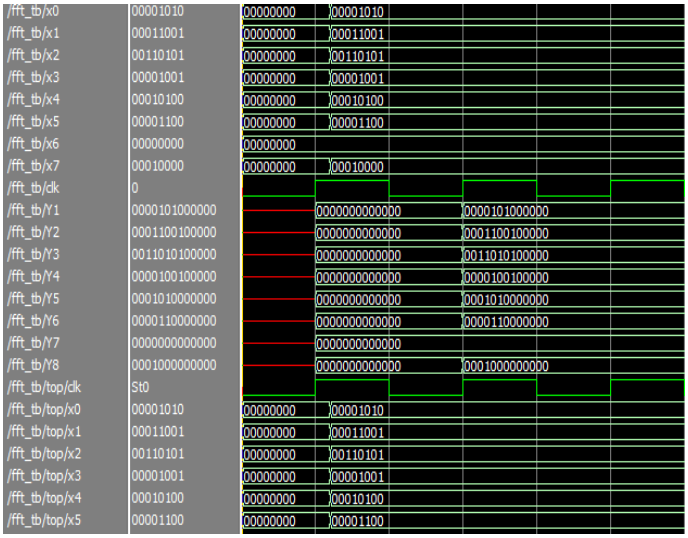


Fig. 4. Simulation result of IFFT.

2) Inverse Fast Fourier Transform (IFFT): The IFFT provides orthogonality between adjacent sub-carriers. Kogge stone adder is used for efficient computation of both IFFT and FFT. DIF algorithm is used to compute IFFT. The simulation result of IFFT is shown in Fig. 4.

3) Addition of Cyclic Prefix: The circular convolution of the signal after IFFT and the channel impulse response occur in the process of addition of CP. Here the end of the OFDM symbol is adjoined at the beginning of the symbol. ICI occurs due to the Doppler shift. Doppler shift is defined as the apparent change in frequency due to the relative motion of the transmitter and the receiver. Due to this effect, the orthogonality of the sub-carriers may get lost. So Cyclic Prefix

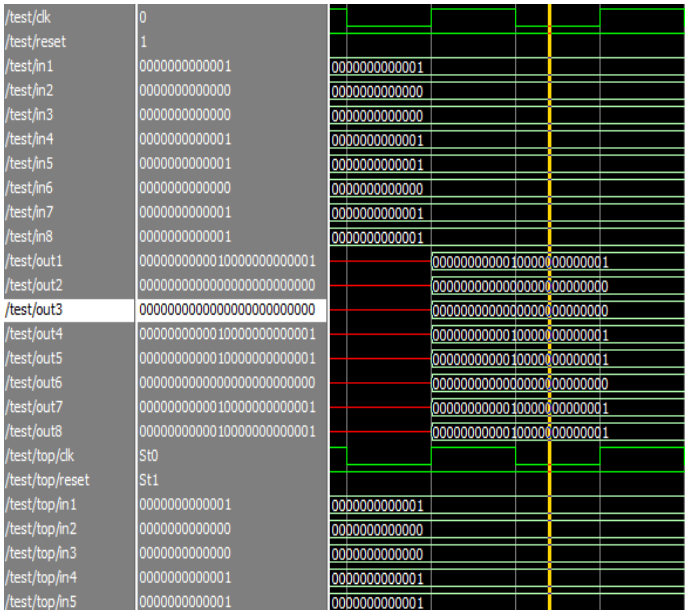


Fig. 5. Simulation result of addition of CP.

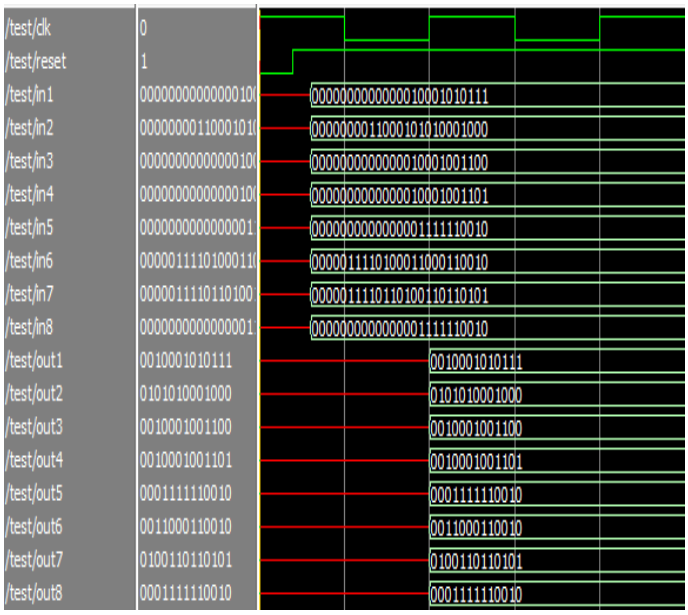


Fig. 6. Simulation result of removal of CP.

is introduced to perpetuate the orthogonality of sub-carriers. It also eliminates Inter-symbol interference. The simulation result of the addition of CP is shown in Fig. 5.

4) Removal of Cyclic Prefix: Eliminating the cyclic prefix is done at the receiver section. Then the signal which is free from cyclic prefix is passed through other receiver blocks. Here, deconvolution of the signal transmitted and the channel impulse response takes place. Therefore, the output of this block is the sampled value of the transmitted signal. The simulation result of the removal of CP is shown in Fig. 6.

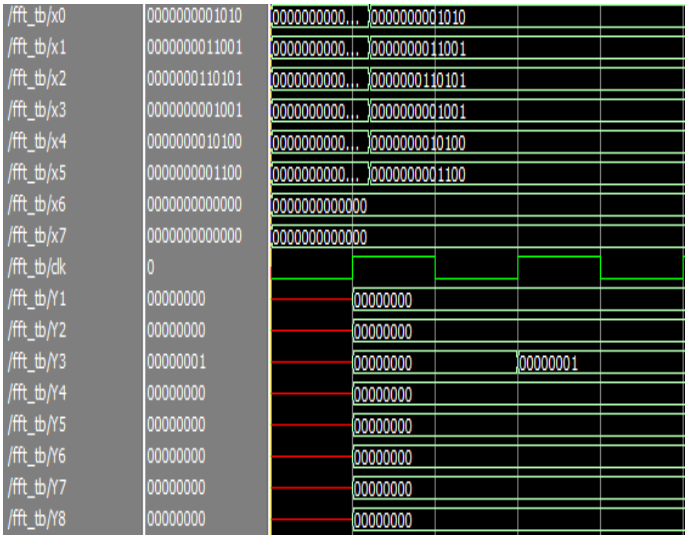


Fig. 7. Simulation result of FFT.

5) Fast Fourier Transform (FFT): The output of the Cyclic prefix removal block is fed as the input of the FFT. For the computation of FFT, DIF algorithm is used. FFT is preferred for faster execution and flexibility. FFT changes the transmitted signal from time domain to frequency domain. This can be implemented in three stages. Each stage consists of four butterflies having upper and lower wing. The input of the first stage is the output of cyclic prefix removal block. The output from the first stage is the input of the second stage and also the output from the second stage is the input of the third stage. The simulation result of FFT is shown in Fig. 7.

B. Matlab Implementation of OFDM and F-OFDM in Terms of PSD and BER

OFDM and F-OFDM have been implemented in MATLAB in terms of the PSD and BER. For F-OFDM, out of band emission is reduced by the design of the filter. Since adjacent channel interference can be reduced effectively using sinc filter, F-OFDM is preferable than OFDM. This is because the impulse response of the filter fades quickly which results in reduced interference between sub-carriers. The F-OFDM handles a sub-band having 50 resource blocks and OFDM handles a band of 600 sub-carriers for the comparison of PSD.

The MATLAB simulation result of OFDM and F-OFDM in terms of PSD as shown in Figs. 8 and 9 respectively. The F-OFDM has lower side lobes than that of OFDM resulting in the enhancement of spectral efficiency. When comparing the PSD plot of both OFDM and Filtered OFDM, it is clear that the Out of Band Radiation (OOB) is reduced effectively for F-OFDM than OFDM. The MATLAB simulation result of OFDM and F-OFDM in terms of BER as shown in Figs. 10 and 11 respectively. When comparing both in terms of BER, both OFDM and Filtered OFDM have approximately equal BER. However in comparison with performance the F-OFDM has good performance than OFDM.

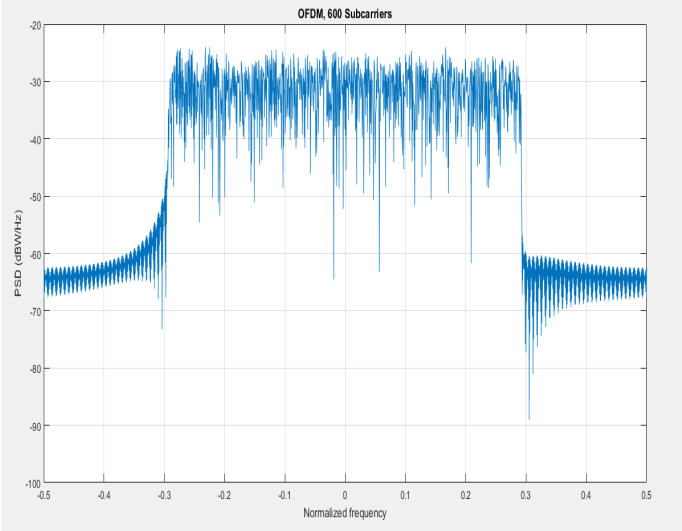


Fig. 8. Simulation result of OFDM in terms of PSD.

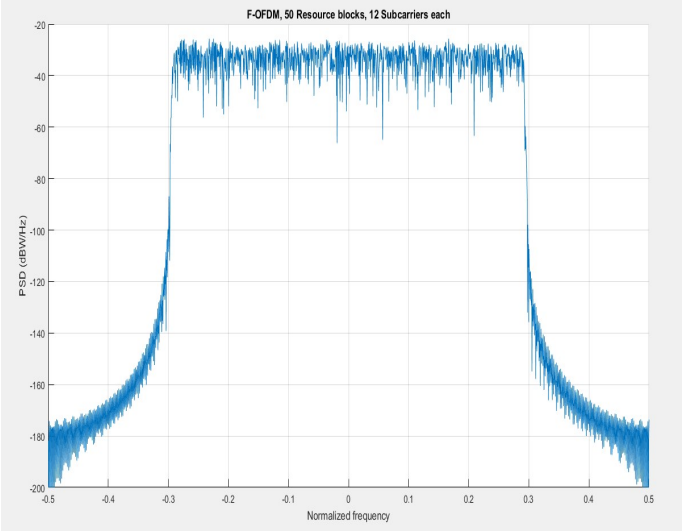


Fig. 9. Simulation result of F-OFDM in terms of PSD.

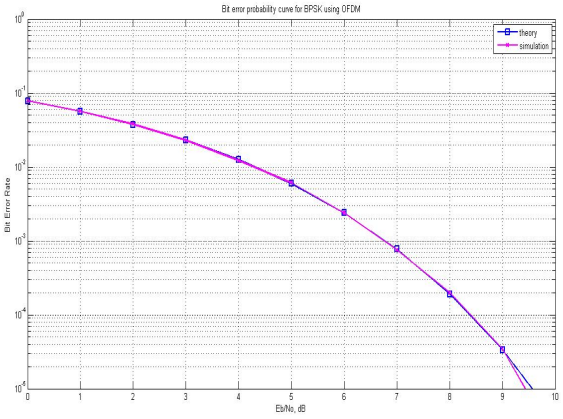


Fig. 10. Simulation result of OFDM in terms of BER.

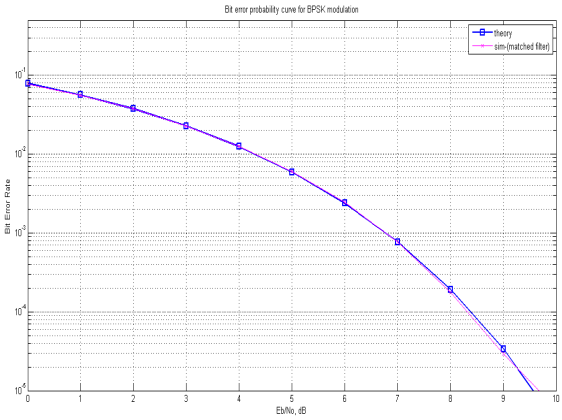


Fig. 11. Simulation result of F-OFDM in terms of BER.

V. Conclusion

OFDM and F-OFDM have been designed using MATLAB and their performance is verified in terms of the Power Spectral Density (PSD) and Bit Error Rate (BER). From the PSD plot, it has been observed that the F-OFDM has lower side lobes than that of OFDM which results in the enhancement of spectral efficiency. In terms of the PSD, the Out of Band Radiation (OOB) is reduced effectively for F-OFDM than OFDM. In order to reduce the OOB radiation, a properly designed sub-band filtering has been done in time domain OFDM symbols. Analysis shows that BER performance of F-OFDM and OFDM gives approximately equal performance. Also, the working of OFDM has been verified in Verilog.

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